



CHAPTER - 14



WAVES

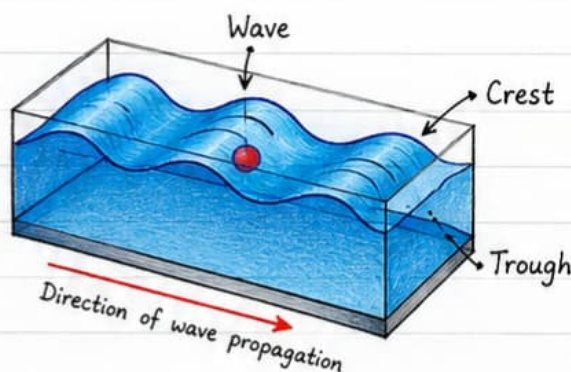
Waves transfer **energy** from one point to another without transferring **matter**.

1. INTRODUCTION

A wave is a **disturbance** that travels through a medium or space, **transferring energy** from one point to another **without** any net transfer of matter.

* Waves are of two types:

- **Mechanical Waves** - require a material medium.
- **Electromagnetic Waves** - do not require any medium.

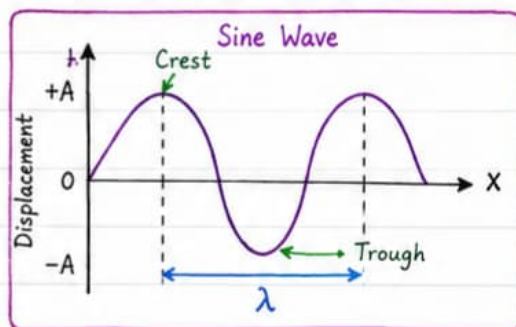


Examples of Waves

- Waves on water surface
- Sound waves in air
- Light waves from the sun
- Radio waves
- Earthquakes (seismic waves)

2. TERMS RELATED TO WAVES

- * **Amplitude (A)** - Maximum displacement of the particle from its mean position.
- * **Wavelength (λ)** - Minimum distance between two successive particles in the same phase.
- * **Time Period (T)** - Time taken by the wave to complete one oscillation.
- * **Frequency (ν)** - Number of oscillations per second.
- * **Wave Speed (v)** - Speed at which the wave travels in the medium.



Important Relations

$$v = f\lambda \quad \text{or} \quad v = \frac{\lambda}{T}$$

$$f = \frac{1}{T}$$

3. TYPES OF WAVES

(A) Mechanical Waves

- Require a material medium for propagation.
- Example: Sound waves, waves on a string, water waves, seismic waves.

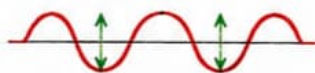
(B) Electromagnetic Waves

- Do not require any medium.
- Can travel in vacuum.
- Example: Light, radio waves, X-rays, γ -rays.

Mechanical Waves are further classified as:

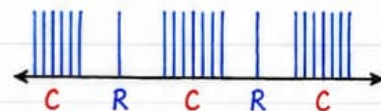
(i) Transverse Waves

Particle vibration is perpendicular to the direction of wave propagation.



(ii) Longitudinal Waves

Particle vibration is parallel to the direction of wave propagation.



Remember

- * Waves carry **energy**, **not** matter.
- * Frequency depends on source, wave speed depends on medium.
- * All electromagnetic waves travel at the speed of light in vacuum ($c = 3 \times 10^8 \text{ m/s}$).

4. WAVE EQUATION AND RELATIONS

The fundamental relation between wave speed (v), frequency (f) and wavelength (λ) is :

$$v = f \lambda$$

Where,

v = speed of wave (m s^{-1})

f = frequency (s^{-1} or Hz)

λ = wavelength (m)

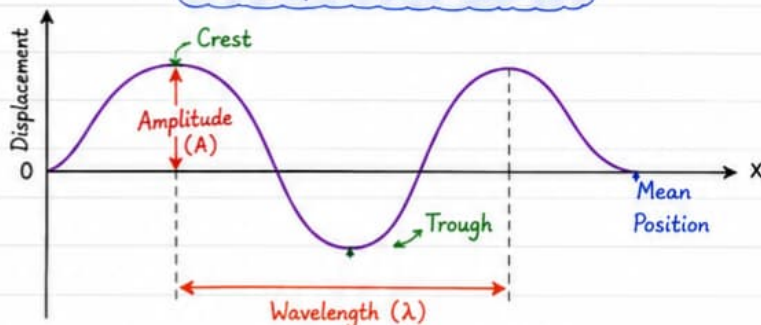
Example 1

A wave has frequency 50 Hz and wavelength 0.20 m. Find its speed.

Solution:

$$v = f \lambda = 50 \times 0.20 = 10 \text{ m s}^{-1}$$

Parts of a Transverse Wave



5. SUPERPOSITION OF WAVES

When two or more waves travel in the same medium, the resultant displacement at any point is the **algebraic sum** of the displacements produced by the individual waves.

* **Constructive interference:** Displacements in the same direction (amplitude increases).

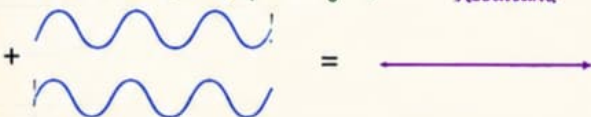
* **Destructive interference:** Displacements in opposite directions (amplitude decreases).

Interference of Two Waves

- Constructive (in phase)



- Destructive (out of phase by π)

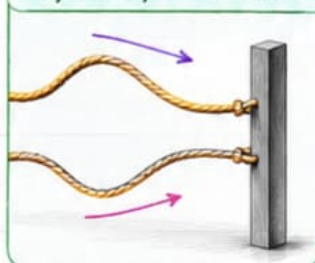


6. REFLECTION OF WAVES

When a wave meets a **boundary**, it returns back into the same medium. Some important points:

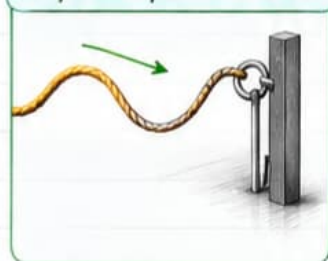
- Angle of incidence (i) = Angle of reflection (r)
 $i = r$
- No change in **frequency**.
- In the same medium, **speed** and **wavelength** remain unchanged.
- A phase change of π (inversion) occurs when reflected from a fixed end.
- No phase change when reflected from a free end.

Reflection from Fixed End



Inversion occurs
(phase change of π)

Reflection from Free End



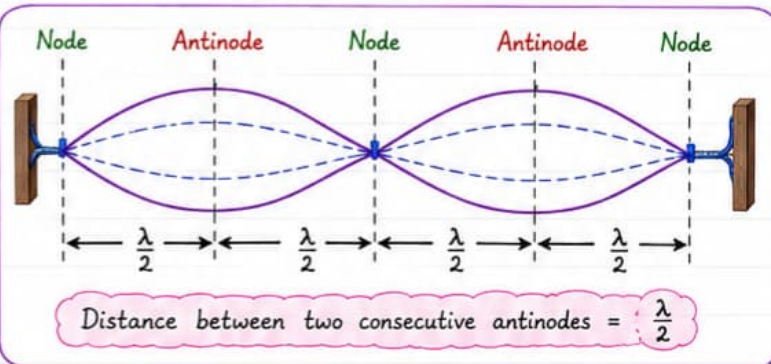
No inversion
(no phase change)

7. STANDING WAVES

When two waves of same frequency and amplitude travel in opposite directions, form a **standing wave**.

- * **Nodes:** Points that always remain at rest (displacement = 0).
- * **Antinodes:** Points of maximum displacement.

Distance between two consecutive nodes = $\frac{\lambda}{2}$



Distance between two consecutive antinodes = $\frac{\lambda}{2}$

Standing waves are formed in strings, air columns in pipes, and many musical instruments.



Keep Practicing

Understand Waves, Understand Nature!

8. DOPPLER EFFECT

Doppler effect is the **apparent change in frequency** of a wave due to **relative motion** between the source and the observer.

(A) Source moving towards stationary observer

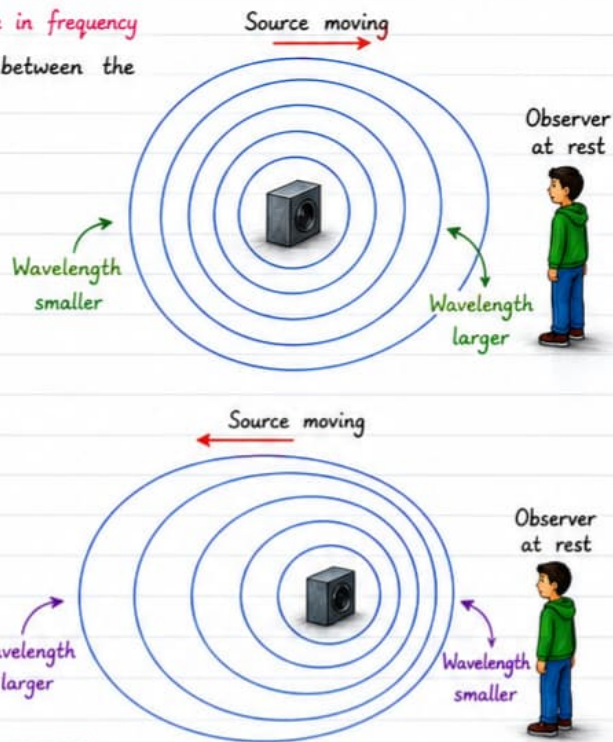
Apparent frequency increases.

$$f' = f \frac{v}{(v - v_s)}$$

(B) Source moving away from stationary observer

Apparent frequency decreases.

$$f' = f \frac{v}{(v + v_s)}$$



Where,

f = actual frequency of source

f' = apparent frequency heard by observer

v = speed of wave in medium

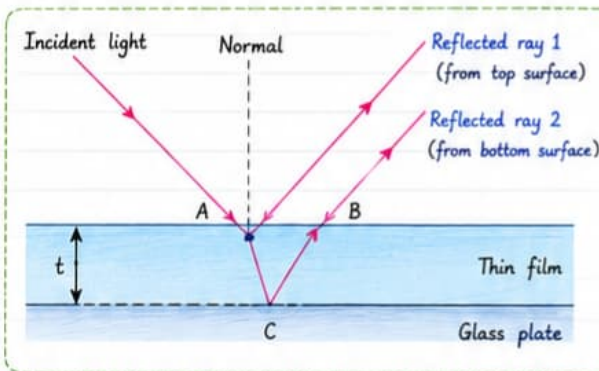
v_s = speed of source

v_o = speed of observer (towards +ve)

* If observer moves towards the source, replace v with $(v - v_o)$ in numerator.

9. INTERFERENCE IN THIN FILMS

When light reflects from a thin film (like soap bubble or oil film), the two reflected rays may interfere with each other to produce **bright or dark bands**.



Condition for Constructive Interference (Bright Fringe)

$$\text{Path difference} = 2\mu t \cos r = n\lambda$$

$$(n = 0, 1, 2, 3, \dots)$$

Condition for Destructive Interference (Dark Fringe)

$$\text{Path difference} = 2\mu t \cos r = \left(n + \frac{1}{2}\right)\lambda$$

$$(n = 0, 1, 2, 3, \dots)$$

Where,

μ = refractive index of film

t = thickness of film

r = angle of refraction in the film

λ = wavelength of light in air

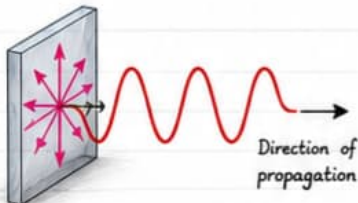
n = order of fringe

10. POLARISATION OF WAVES

Polarisation is the phenomenon in which the vibrations of a **transverse wave** are restricted to **one particular direction** perpendicular to the direction of propagation.

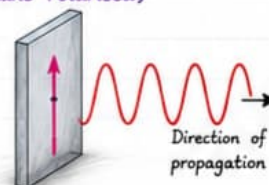
Unpolarised Light

* Vibrations in all possible directions perpendicular to propagation.



Polarised Light (Plane Polarised)

* Vibrations in only one direction (vertical in this case).



Applications

- Polaroid glasses
- Photography
- 3D movies
- Reduction of glare

11. IMPORTANT POINTS (QUICK REVISION)

- * Waves transfer **energy**, not matter.
- * **Mechanical waves** need a medium.
- * **Electromagnetic waves** can travel in vacuum.
- * $v = f\lambda$ is a fundamental relation.
- * **Reflection**: angle of incidence = angle of reflection.
- * **Standing waves** have nodes and antinodes.



- * **Interference** can be constructive or destructive.
- * **Doppler effect** depends on relative motion.
- * **Polarisation** is possible only in transverse waves.
- * The speed of light in vacuum, $c = 3 \times 10^8$ m/s.
- * **Practice numerical problems for better understanding!** 😊

→ ~~~~~ → Waves are all around us - from raindrops to radio waves! < ~~~~~ <

12. ENERGY TRANSPORT IN WAVES

- Waves carry **energy** from one place to another without any **net transport of matter**.
- The average power (P_{avg}) transmitted by a sinusoidal wave on a stretched string is :

$$P_{avg} = \frac{1}{2} \mu \omega^2 A^2 v$$

Where,

μ = mass per unit length of the string (kg m^{-1})

ω = angular frequency = $2\pi f$ (rad s^{-1})

A = amplitude of the wave (m)

v = speed of the wave (m s^{-1})

Energy moves...

Matter doesn't!

13. EXAMPLES

Example 1: Wave on a String

A sinusoidal wave on a string is described by

$$y(x, t) = 0.02 \sin(\pi x - 10\pi t)$$

Find (i) amplitude (ii) wavelength
(iii) frequency (iv) wave speed.

Solution:

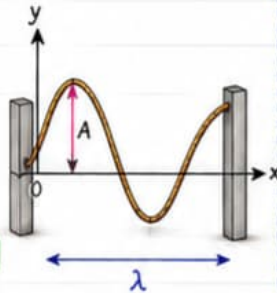
Comparing with $y = A \sin(kx - \omega t)$

(i) $A = 0.02 \text{ m}$

(ii) $k = \pi = \frac{2\pi}{\lambda} \rightarrow \lambda = 2 \text{ m}$

(iii) $\omega = 10\pi = 2\pi f \rightarrow f = 5 \text{ Hz}$

(iv) $v = f\lambda = 5 \times 2 = 10 \text{ m s}^{-1}$



Example 2: Sound Wave in Air

A sound wave in air has frequency **340 Hz** and speed 340 m s^{-1} . Find its wavelength.

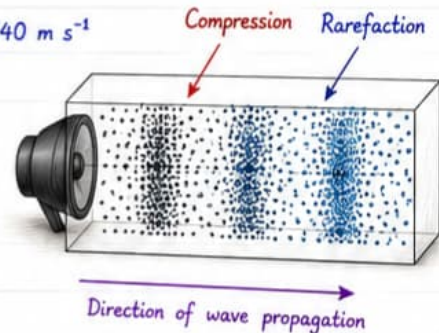
Solution:

Given:

$$f = 340 \text{ Hz}, v = 340 \text{ m s}^{-1}$$

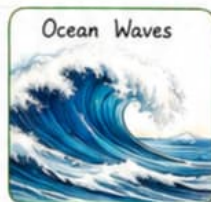
We know, $v = f\lambda$

$$\lambda = \frac{v}{f} = \frac{340}{340} = 1 \text{ m}$$

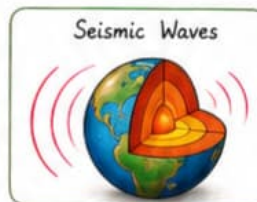


14. WAVES IN NATURE AND TECHNOLOGY

- Ocean waves shape the sea beaches.
- Seismic waves help in studying the Earth's interior.
- Radio waves enable communication.
- Ultrasound is used in medical imaging.
- Light waves make vision possible.



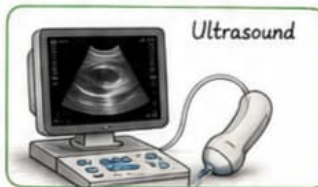
Ocean Waves



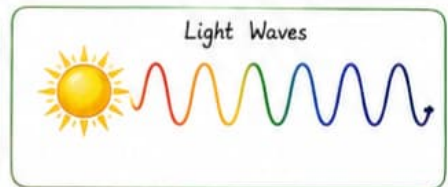
Seismic Waves



Radio Waves



Ultrasound



Light Waves

SUMMARY

- Waves transfer **energy**, not matter.
- $v = f\lambda$ is the fundamental relation.
- Waves show **reflection**, **interference**, **diffraction**, **polarisation** and **Doppler effect**.
- Standing waves have nodes and antinodes.
- Different types of waves occur in nature and have practical applications.



KEY FORMULAS (AT A GLANCE)

$$v = f\lambda$$

$$v = \frac{\lambda}{T}$$

$$f = \frac{1}{T}$$

$$P_{avg} = \frac{1}{2} \mu \omega^2 A^2 v$$

$$\omega = 2\pi f$$

$$\text{Constructive: } \Delta = n\lambda$$

$$\text{Destructive: } \Delta = \left(n + \frac{1}{2}\right) \lambda$$

$$\Delta = 2\mu t \cos r$$

(Interference in thin films)

$$\text{Doppler (towards): } f' = f \cdot \frac{v}{v - v_s}$$

$$\text{Doppler (away): } f' = f \cdot \frac{v}{v + v_s}$$

REMEMBER



✓ Understand concepts clearly.

✓ Draw diagrams to remember better.

✓ Practice numerical problems.

✓ Revise regularly!

All the Best!

